

CMU-CS 462: Software Measurement and Analysis

Spring 2025-2026

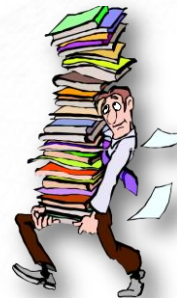
Review (Ôn tập)

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1

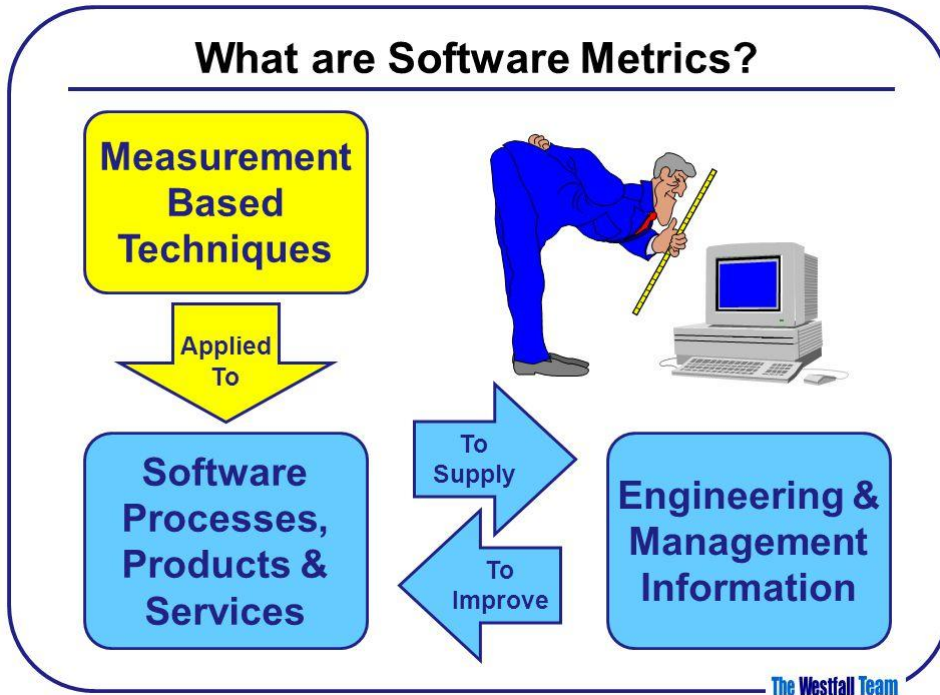
Contents

1. Overview of Software Metrics
2. GQM
3. Size (LOC, FP, UCP)
4. COCOMO II
5. Test Metrics
6. Quality Metrics
7. EVM



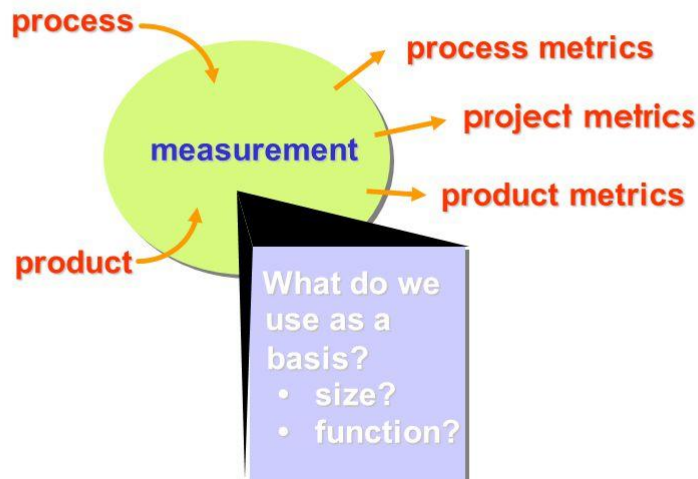
2

What are Software Metrics?



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What Should Be Measured?



6

4

Two Problem Categories

■ Components of a measurement system:

$m = \langle \text{attribute, scale, unit} \rangle$

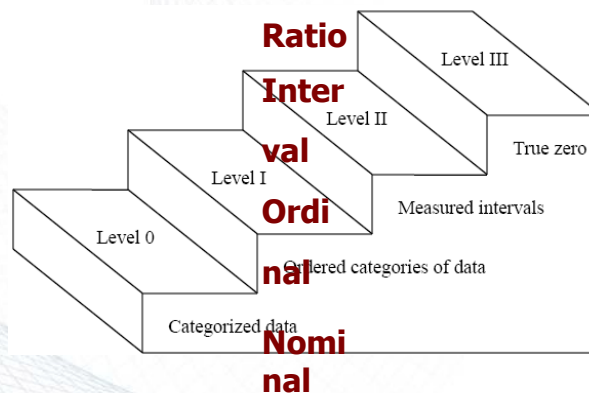
- Attribute is what is being measured (e.g., size of a program)
- Scale is the standard and scope of measurement (e.g., nominal, ordinal, ratio scale, etc.)
- Unit is the physical meaning of scale (e.g., a positive integer, a symbol, etc.)
- Determining the value of an attribute of an entity.
- Determining the class of entities to which the measurement relates.

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5

5

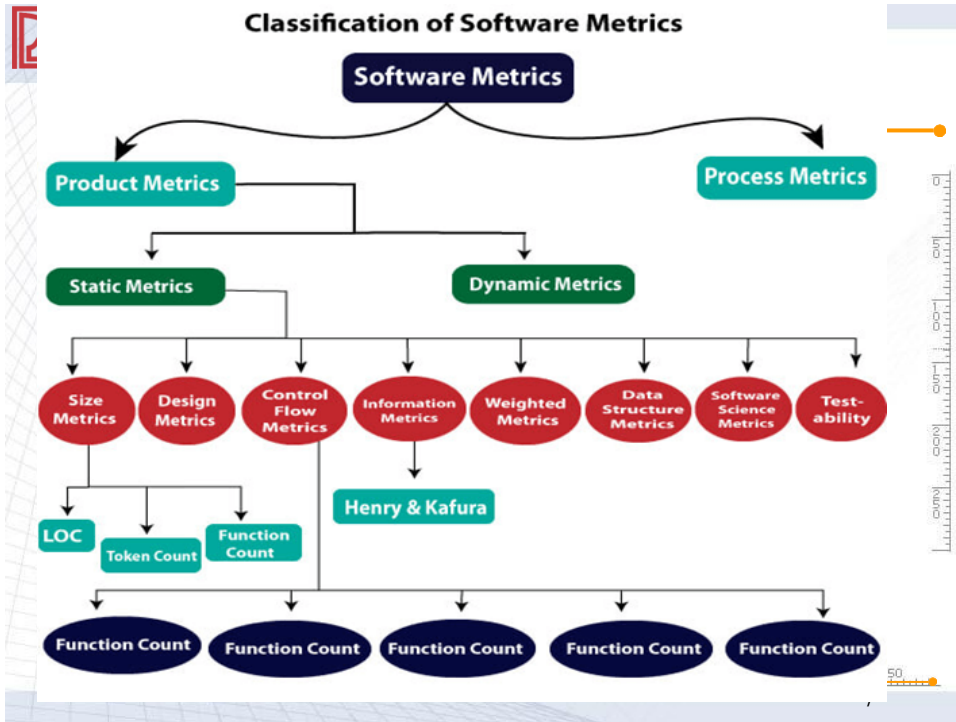
Scales (Summary)



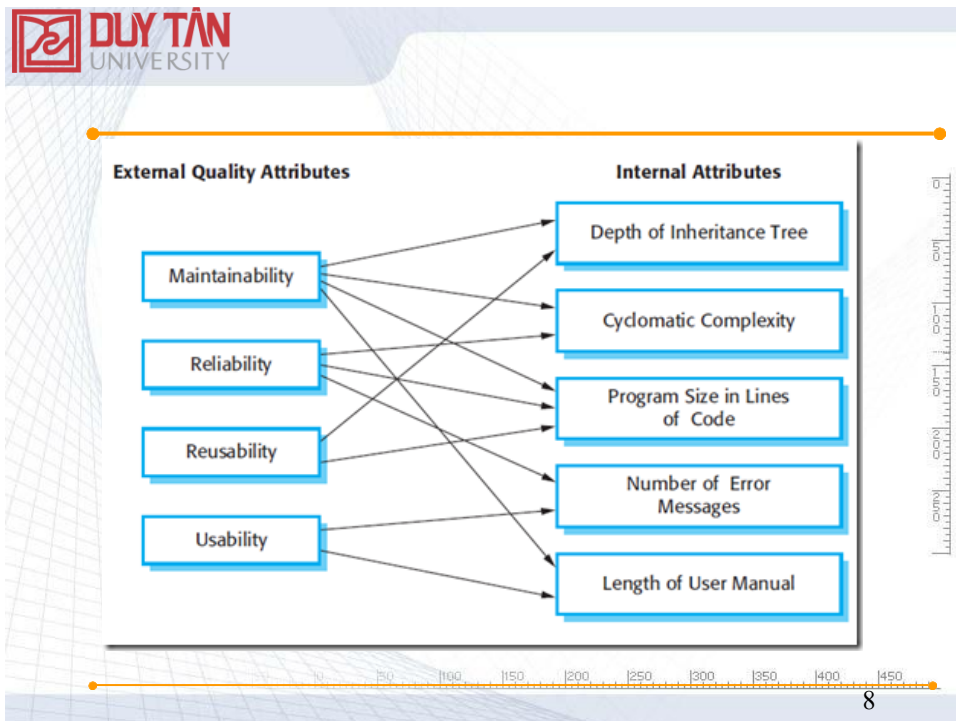
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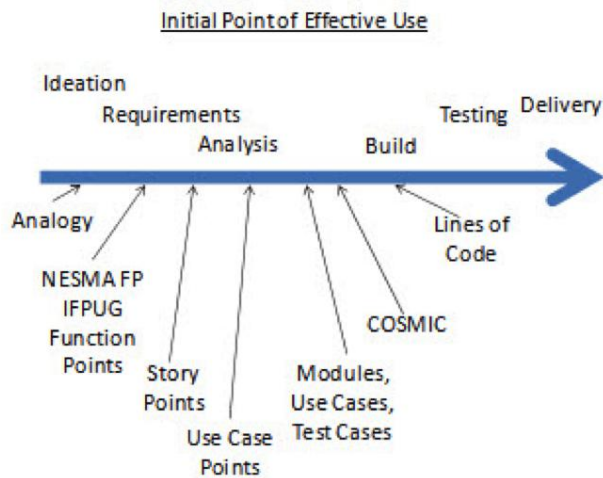
Goal: evaluate the effectiveness of writing sw code using standards

Questions: who's using standard? What is coders experience? What is LOC quality?

Metrics:

- Proportions of "coders"
 - using std
 - by Progr. Lang.
- Experience of "coders"
 - with std
 - with PL
 - by env
 - ...
- Sw size/length
 - LOC
 - FP/CFP
 - ...
- Effort
- Errors

Metrics methods



Function Point

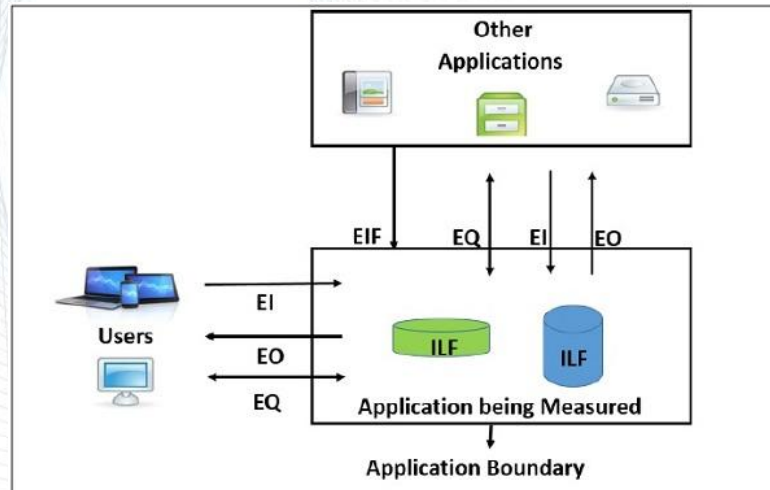
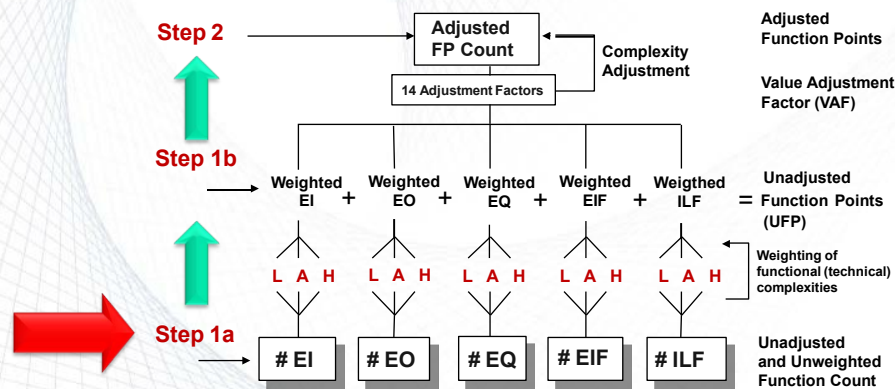


Figure 1: Application Boundary, Data Functions, Transaction Functions

11

11

FP Counting Summary



12/01/2026

12

12



USC COCOMO II Model

USC COCOMO II, 2000.0 - C:\Cocomo\Examples\TPS.est

File Edit View Parameters Calibrate Phase Maintenance Help

Project Name: **TPS** Scale Factor: Schedule: Development Model: **Post Architecture**

K	Module Name	Module Size	LABOR Rate (\$/month)	EAF	Language	NOM Effort DEV	EST Effort DEV	PROD	COST	INST COST	Staff	RISK
	System-Reuse	A:1900	0.00	0.60	C	6.5	3.9	484.0	0.00	0.0	0.3	0.0
	System-New	S:18000	0.00	0.60	C	61.9	37.2	484.0	0.00	0.0	2.5	0.0
	Applications	S:800	0.00	0.60	C	2.8	1.7	484.0	0.00	0.0	0.1	0.0
	Diagnosis	F:24000	0.00	0.60	Third Generat	82.5	49.6	484.0	0.00	0.0	3.4	0.0

	Estimated	Effort	Sched	PROD	COST	INST	Staff	RISK
Total Lines of Code: 44700	Optimistic	73.9	13.7	604.9	0.00	0.0	5.4	
	Most Likely	92.4	14.7	484.0	0.00	0.0	6.3	0.0
	Pessimistic	115.5	15.7	387.2	0.00	0.0	7.3	

Ready

6/25/2015

13



COCOMO /2

- The original COCOMO is a collection of three models:
 - Basic model** that is applied early in the project
 - Intermediate model** that is applied after requirements acquisition
 - Advanced model** that is applied after design is complete
- All three models take the form:

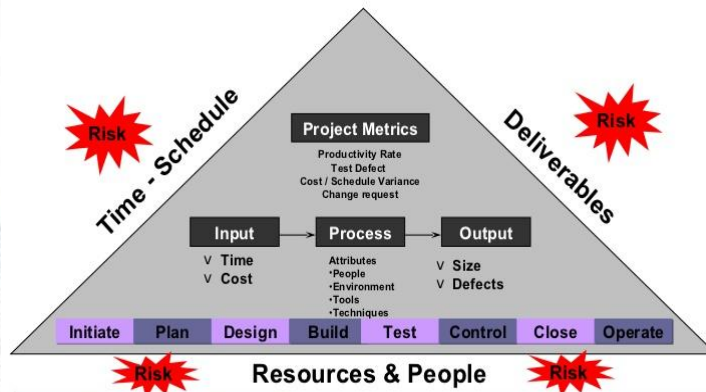
$$E = aS^b \times EAF \quad T_{dev} = cE^d$$

where

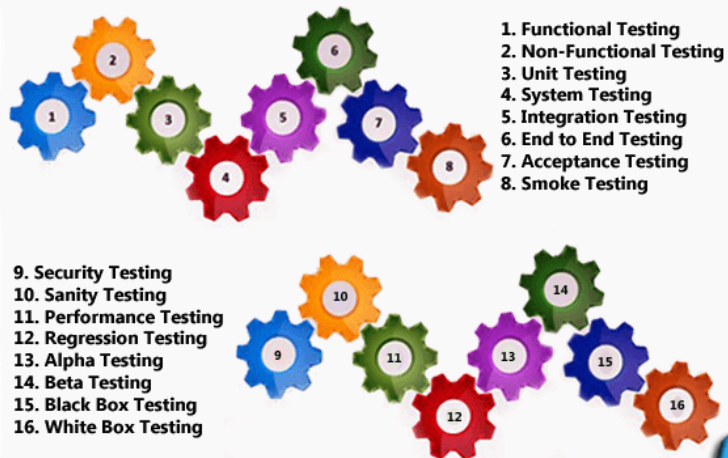
- E is effort in person months
- T_{dev} is the development time
- S is size measured in KLOC
- EAF is an effort adjustment factor (1 in the Basic model)
- Factors a , b , c and d depend on the **development mode**.

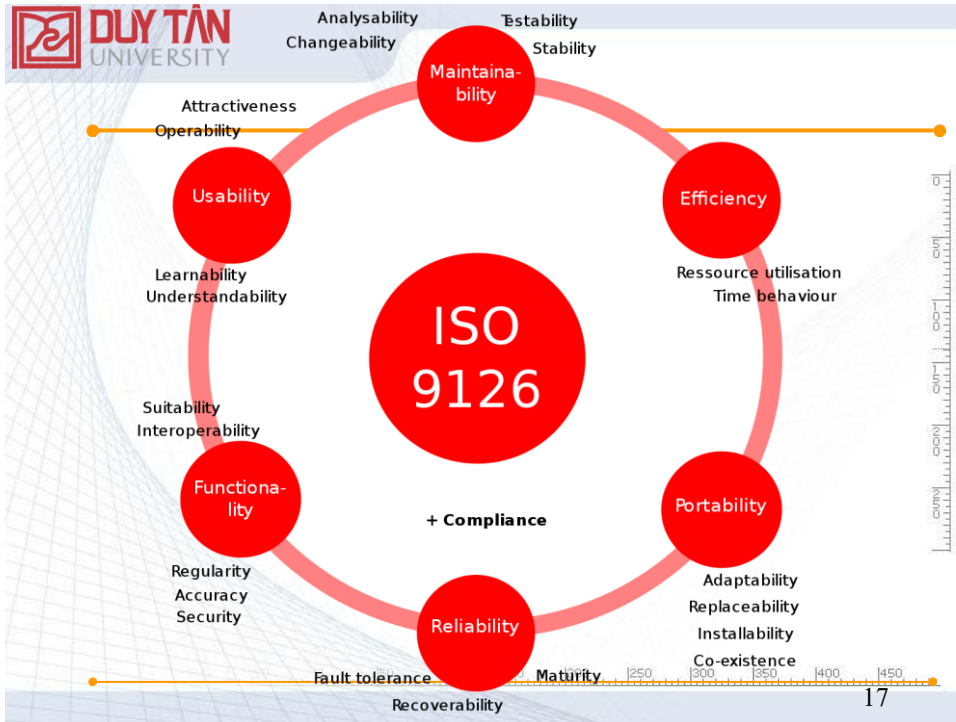
14

Project Process Metrics Model

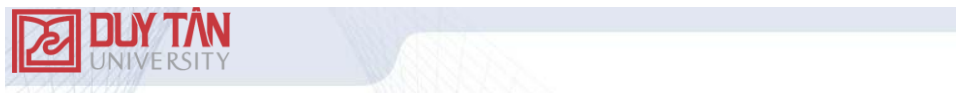


Types of Software Testing

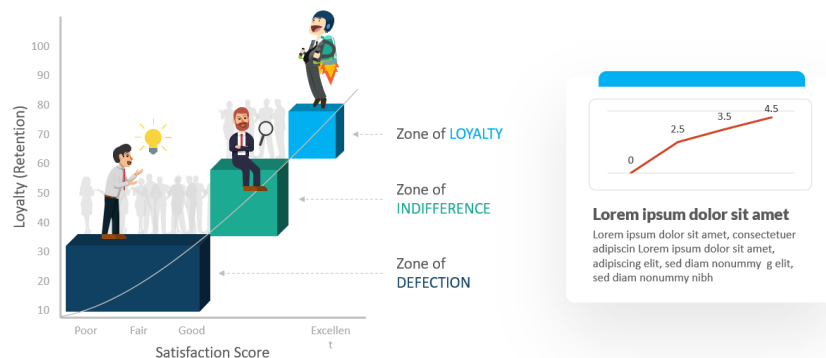




17



CUSTOMER SATISFACTION METRICS



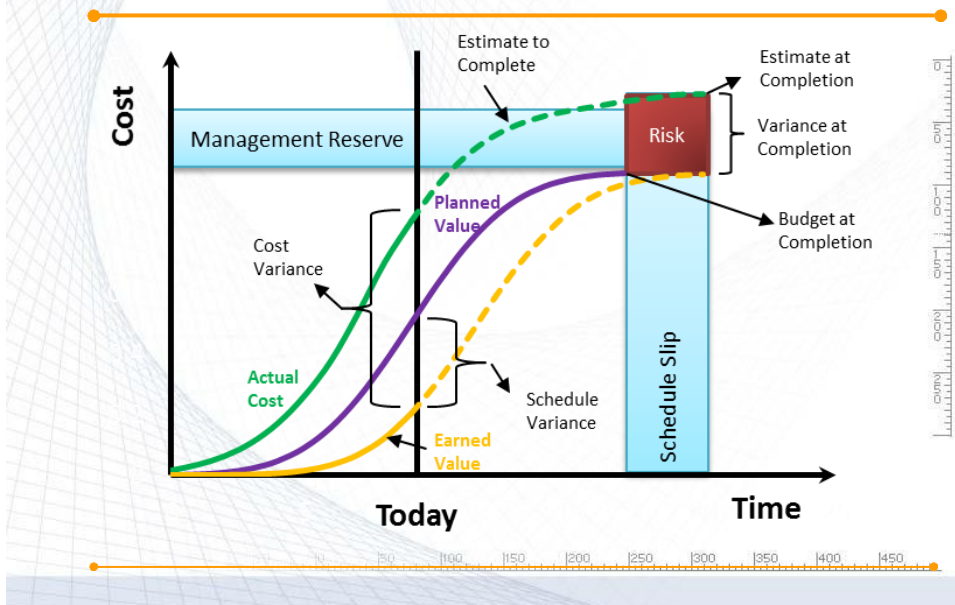
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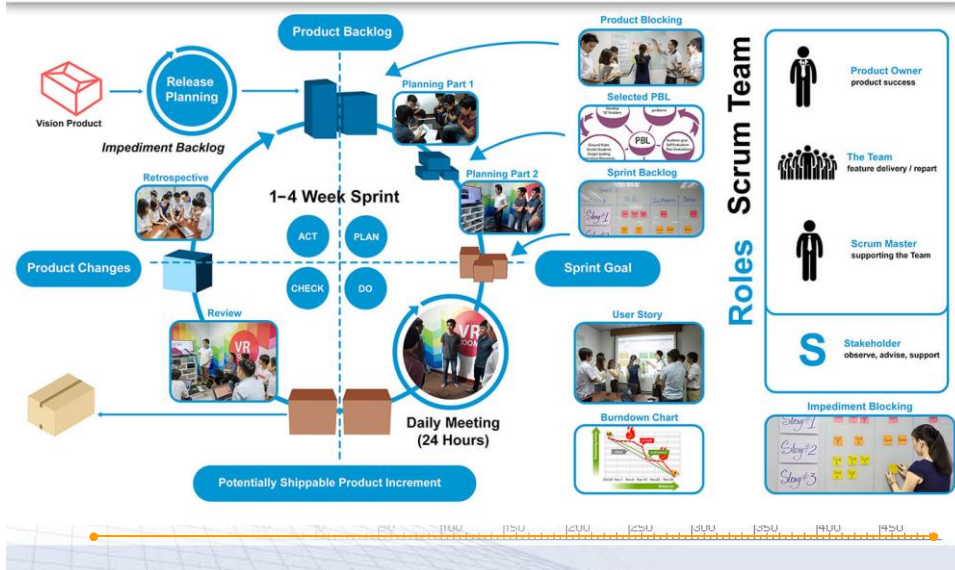


18



19

AGILE SCRUM PROCESS



20

Nội dung thi

- **Cấu trúc đề thi:**
 - 15 câu MCQs (đọc lại toàn bộ Slides) → 3đ
 - 10 câu SAs (đọc lại toàn bộ slides và tài liệu tham khảo) → 3đ
 - Chỉ trả lời từ 1 – 2 sentences
 - Các câu hỏi tập trung cho việc giải thích, nêu nội dung, liệt kê
 - 01 Bài tập phân tích và tính FP, UCP và EVM (tính tại một thời điểm bất kỳ) → 4 đ
 - Xem lại các dạng bài tập đã làm, các ví dụ đã trình bày
 - Chú ý các công thức tính, cách tính, cách trình bày đã được học
 - Thời lượng 90'

21

Nội dung thi

- **Cấu trúc đề thi:**
- 01 Bài tập phân tích và tính FP, UCP và EVM (tính tại một thời điểm bất kỳ) → 4 đ
 - Phân tích, nhận diện và xác định số EI, EO, EQ, ILF, EIF; với Fi cho trước → tính FP
 - Tương tự xác định UUAW, UUCW, với Fi cho trước → tính UCP
 - Từ FP, UCP tính effort cho dự án
 - Cho dữ liệu theo bảng, tính các chỉ số, giá trị thu được tại tuần/ngày/tháng thứ x. xác định các ước tính và các kết luận tương ứng
- Thời lượng 90'

22

Group discussion

What have you learned through this course?

Let's discuss in groups of 5 students, then each group write down keywords representing the content, issues that you have learned, the group that writes more keywords will get plus points for the whole group!

Discuss for 20 minutes.

23

Q & A- Discussion

Questions/Discussion



24

References

Standards:

- [IEEE730] IEEE 730-2002: IEEE Standard for Software Quality Assurance Plans, Institute of Electrical and Electronics Engineers, 29-Sep-2002.
- [ISO9126] ISO/IEC 9126-2001: Software engineering -- Product quality -- Parts 1-4, ed. 1, Current stage 90.20 JTC 1/SC 7, 2001.

Software Quality:

- [Kan02] Stephen H. Kan: Metrics and Models in Software Quality Engineering, 2nd Edition, Addison-Wesley, 2002.

- <https://www.apptio.com/solutions/it-metrics-kpis/?wvideo=okc05zis0g> (Review)

- <https://www.youtube.com/watch?v=lc0Y1BX7Fig>